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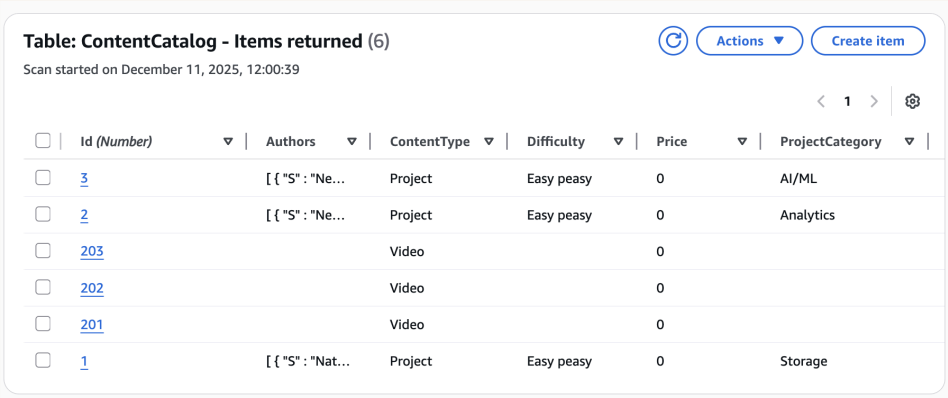
Load Data into a DynamoDB Table

CH Arpita Chowdhury



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Introducing Today's Project!

What is Amazon DynamoDB?

Amazon DynamoDB is a fully managed NoSQL database service that provides fast, predictable performance and automatic scalability. It stores data in key value and document formats and removes the need to manage servers, replication, backups, or hardware provisioning.

How I used Amazon DynamoDB in this project

In today's project, I used Amazon DynamoDB to create a table with Amazon DynamoDB, upload data into DynamoDB using AWS CloudShell and edit data in DynamoDB tables.

One thing I didn't expect in this project was...

One thing I didn't expect in this project is encounter error while trying to view data in the table.

This project took me...

This project took me 2 hours.

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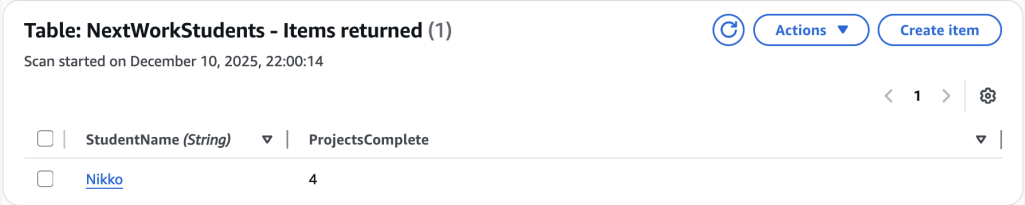
Create a DynamoDB table

DynamoDB tables organize data using primary keys for item uniqueness, partitions for physical storage distribution, and indexes (LSIs and GSIs) for alternative query patterns.

An attribute is a single piece of data that describes an item.



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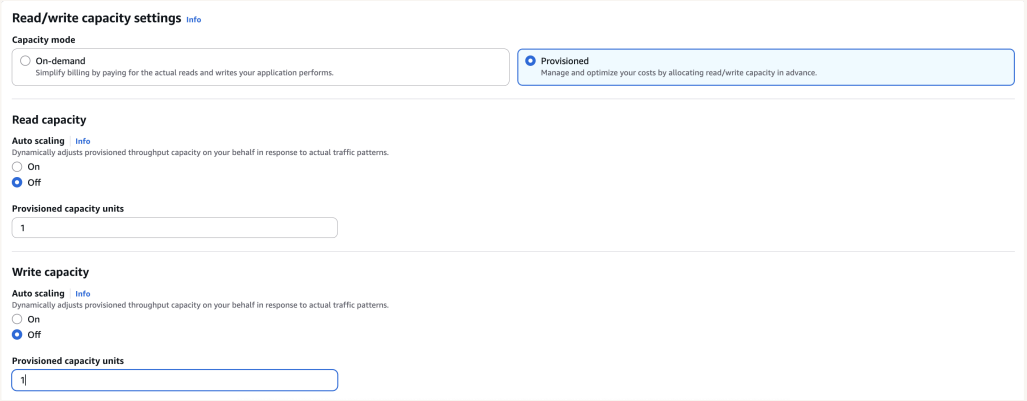
Read and Write Capacity

Read capacity units (RCUs) and write capacity units (WCUs) are the units that determine how much read and write capacity a DynamoDB table can handle, based on item size and consistency requirements.

Amazon DynamoDB's Free Tier covers 25 GB of storage, plus up to 25 RCUs and 25 WCUs each month at no cost. I turned off auto scaling because I wanted to control capacity manually and avoid unexpected throughput increases that could raise costs.



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Using CLI and CloudShell

AWS CloudShell is a browser-based shell that provides a pre-configured Linux environment with the AWS CLI and tools already installed, allowing you to run commands against your AWS resources without local setup.

AWS CLI is a tool that provides programmatic access to AWS APIs through terminal commands, supporting automation, scripting, and resource management.

I ran a CLI command in AWS CloudShell that created tables.

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Loading Data with CLI

I ran a CLI command in AWS CloudShell that retrieved a ZIP file of sample data, extracted it, and then populated my DynamoDB tables by running batch-write-item for each JSON dataset.

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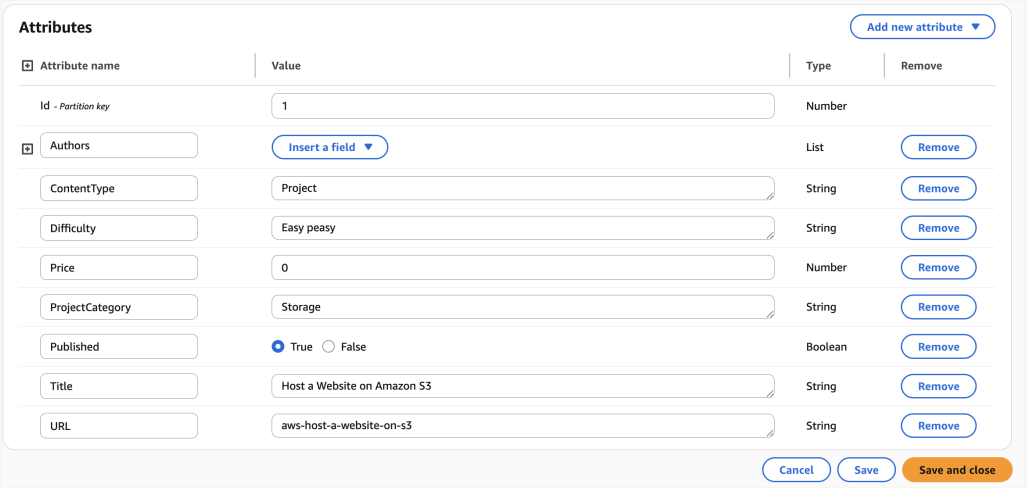
Observing Item Attributes



I checked a ContentCatalog item, which had the following attributes: Id, Author, ContentType, Difficulty, Price, ProjectCategory, Published, Services, Title and URL.

I checked another ContentCatalog item, which had a different set of attributes ContentType, price, Services, Title, URI and VideoType.

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Benefits of DynamoDB

A benefit of DynamoDB over relational databases is flexibility, because unlike relational databases that require rigid table schemas and predefined columns, DynamoDB allows flexible data models where attributes can be added or omitted on a per-item basis, making it easier to evolve applications quickly.

Another benefit over relational databases is speed, because it automatically distributes data across partitions and uses optimized key-value access patterns, so reads and writes are extremely fast compared to relational databases that rely on complex joins and indexing.

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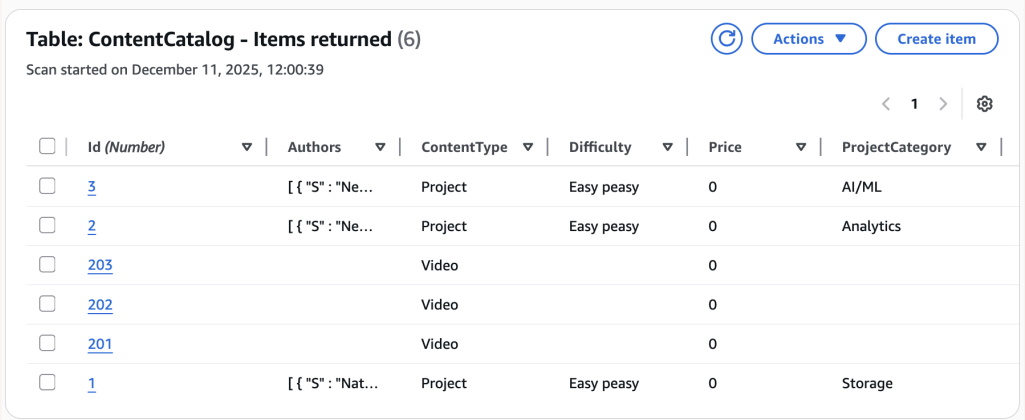

Table: ContentCatalog - Items returned (6)

Scan started on December 11, 2025, 12:00:39

Actions | Create item

1

Id *(Number)*	Authors	ContentType	Difficulty	Price	ProjectCategory
3	[{ "S" : "Ne...	Project	Easy peasy	0	AI/ML
2	[{ "S" : "Ne...	Project	Easy peasy	0	Analytics
203		Video		0	
202		Video		0	
201		Video		0	
1	[{ "S" : "Nat...	Project	Easy peasy	0	Storage

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